

**BACHELOR OF TECHNOLOGY**  
**ELECTRICAL ENGINEERING**  
**THIRD YEAR (SIXTH SEMESTER)**  
**W.E.F. ADMISSION BATCH 2023-24**

| Sl. No.                         | Category | Course Code | Course                               | Contact Hrs. L-T-P | Credit | University Marks | Internal Evaluation |
|---------------------------------|----------|-------------|--------------------------------------|--------------------|--------|------------------|---------------------|
| Subject (Theory)                |          |             |                                      |                    |        |                  |                     |
| 1                               | PC       | EEPC3004    | Power System - II                    | 3-0-0              | 3      | 100              | 50                  |
| 2                               | PC       | EOPC3002    | Digital Signal Processing            | 3-0-0              | 3      | 100              | 50                  |
| 3                               | PE       | EEPE3005    | Renewable Energy Sources             | 3-0-0              | 3      | 100              | 50                  |
|                                 |          | EEPE3006    | Power Station Engineering            |                    |        |                  |                     |
|                                 |          | EEPE3007    | Electric Vehicle                     |                    |        |                  |                     |
|                                 |          | ETPE3008    | Industrial Power Electronics         |                    |        |                  |                     |
|                                 |          | -           | -                                    |                    |        |                  |                     |
|                                 |          | -           | -                                    |                    |        |                  |                     |
| 4                               | PE       | EEPE3009    | Power System Protection              | 3-0-0              | 3      | 100              | 50                  |
|                                 |          | EEPE3010    | High Voltage Engineering             |                    |        |                  |                     |
|                                 |          | EEPE3011    | Electrical Power Quality             |                    |        |                  |                     |
|                                 |          | EEPE3012    | Storage Technology                   |                    |        |                  |                     |
|                                 |          | -           | -                                    |                    |        |                  |                     |
|                                 |          | -           | -                                    |                    |        |                  |                     |
|                                 |          | -           | -                                    |                    |        |                  |                     |
| 5                               | HS       | HSHS3003    | Professional Ethics                  | 3-0-0              | 2      | 100              | 50                  |
|                                 |          | HSHS3002    | Entrepreneurship Development         |                    |        |                  |                     |
| 6                               | MC       | MCMC3002    | Industrial Safety Engineering        | 3-0-0              | 2      | 100              | 50                  |
|                                 |          | MCMC3001    | Environmental Engineering            |                    |        |                  |                     |
| Subject (Sessional / Practical) |          |             |                                      |                    |        |                  |                     |
| 7                               | PSI      | CSPS3202    | Project for Product Development - I  | 0-0-6              | 3      | -                | 100                 |
| 8                               | PC       | EEPC3204    | Power System Laboratory              | 0-0-3              | 1.5    | -                | 100                 |
| 9                               | PC       | EEPC3205    | Digital Signal Processing Laboratory | 0-0-3              | 1.5    | -                | 100                 |
|                                 |          |             | Total                                | 18-0-12            | 22     | 600              | 600                 |

**NB : Minimum 4 weeks of Summer Internship and Research Experience-II (SIRE-II) after 6th Semester during the vacation.**

[Click here to view/download the syllabus of the subjects.](#)

## EEPC3004 POWER SYSTEM - II (3-0-0)

**Course Learning Objectives:** This course will enable students to:

- Impact knowledge on need for operational studies, and To model the power system under steady state operating condition.
- Understand and apply iterative techniques for power flow analysis.
- Understand power system economics fundamentals and generation allocation
- Understand automatic generation control in power system
- Model of carry out short circuit studies for power system during symmetrical and fault
- Study about the various methods for analyzing power system stability

### Module I (7 Hours)

Load Flow Studies: Representation of Power Systems Components, Complex Power, Concept of real and reactive powers, Per-Unit System, Single Line or One Line Diagram and Reactance Diagrams, Formation of Y-bus matrix, Z-Bus Matrix, Network model formulation, Load flow problem, Gauss Siedel and Newton Raphson method for power flow, fast decoupled load flow, Regulating Transformer.

### Module II (6 Hours)

Economic Operation of Power System: Formulation of economic dispatch problem, I/O cost characterization, Incremental cost curve, Coordination equations without and with loss (No derivation of loss coefficients), Solution by Direct method and  $\lambda$ -iteration method, Statement of unit commitment problem, Priority-list method.

### Module III (6 Hours)

Automatic Generation Control: P-F versus Q-V control, Load frequency control(single Area), Turbine speed governing system, Modeling of Speed Governing system, Turbine, Generator, Steady states analysis, dynamic response, control Area, Load frequency control and economic dispatch, Two area load frequency control.

### Module IV (6 Hours)

Fault Analysis : Symmetrical Faults on 3-phase system, Transient on a transmission line, Synchronous machine (no load and load), Short circuit current and MVA, Symmetrical Component transformation (positive, negative and zero sequences), Representation of generators, lines and transformers in sequence networks, Unsymmetrical fault analysis- LG, LL, LLG, Open conductor fault

### Module V (5 Hours)

Power system stability: Power angle equation, steady state stability, transient stability, equal area criterion, factor affecting transient stability

**Course Outcomes (CO):** On completion of this course, students are able to:

- CO1. Compute load flow solution by using different techniques.
- CO2. Determine the economical load distribution between the generating station without and with transmission losses.
- CO3. Analyze automatic generation control schemes in single and two area power system.
- CO4. Evaluate Fault currents for symmetrical and unsymmetrical types of fault.
- CO5. Understand concept of power system stability

### Text Book(s):

1. J. J. Grainger and W. D. Stevenson, Power System Analysis. New York, NY, USA: McGraw-Hill Education, 1994.
2. C. L. Wadhwa, Electrical Power Systems, 6th ed. New Delhi, India: New Age International Publishers, 2010.
3. D. P. Kothari and I. J. Nagrath, Modern Power System Analysis, 4th ed. New Delhi, India: McGraw-Hill Education, 2011.

**Reference Book(s):**

1. O. I. Elgerd, Electric Energy Systems Theory. New York, NY, USA: McGraw-Hill Education, 1995.
2. B. M. Weedy, B. J. Cory, N. Jenkins, J. Ekanayake, and G. Strbac, Electric Power Systems, 5th ed. Hoboken, NJ, USA: Wiley, 2012.
3. R. D. Bergen and V. Vittal, Power System Analysis. Upper Saddle River, NJ, USA: Pearson Education Inc., 1999.
4. M. L. Soni, P. V. Gupta, U. S. Bhatnagar, & A. Chakrabarti, "A Textbook on Power System Engineering", Dhanpat Rai & Co., 1998.
5. NPTEL - <https://nptel.ac.in/courses/108/105/108105067/>

## EEPC3204 POWER SYSTEM LABORATORY (0-0-3)

**Course objectives:** This course aims to

- Understand and apply iterative techniques for power flow analysis.
- Understand power system economics and generation allocation
- Understand automatic Load frequency control in power system
- Carry out short circuit studies for power system during symmetrical and unsymmetrical fault
- Study IDMT Over current relay, MHO relay, Over / under voltage relay

### List of Experiments

#### Simulation Based Experiment

1. Compute Power Flow Solution by using different techniques like GS, NR and FDLF Method
2. Economic Dispatch In Power Systems
3. Load – Frequency Control Of Single and Two Area Power Systems
4. To perform symmetrical and unsymmetrical fault analysis in a power system.
5. Transient stability Analysis in a SMIB & Multi machine systems.

#### Hardware Based Experiment

1. To study the IDMT over-current relay and with different plug setting and time setting multipliers and plot its time – current characteristics ( Numerical / Electromechanical)
2. To determine the operating characteristics of biased different relay with different % of biasing ( Numerical / Electromechanical)
3. Over/Under Voltage Relay ( Numerical / Electromechanical)
4. To study the MHO and reactance type distance relays. ( Numerical / Electromechanical)
5. To study Over/Under Frequency Relay

**Course Outcomes (CO):** On completion of this course, students are able to:

- CO1. Compute load flow solution by using different techniques.
- CO2. Determine the economical load distribution between the generating station.
- CO3. Analyze automatic generation control schemes in single and two area power system.
- CO4. Evaluate Fault currents for symmetrical and unsymmetrical types of fault.
- CO5. Understand concept of power system stability.
- CO6. Understand different power system protection scheme

#### Text Book(s):

1. C. L. Wadhwa, Electrical Power Systems, 6th ed. New Delhi, India: New Age International Publishers, 2010.
2. D. P. Kothari and I. J. Nagrath, Modern Power System Analysis, 4th ed. New Delhi, India: McGraw-Hill Education, 2011.
3. Protection and Switchgear - B.Bhalja, R.P.Maheshwari, N.G. Chothani, OXFORD University Press.
1. Power System Protection and Switchgear - Badri Ram, Vishwakarma, Tata McGraw hill.

#### Reference Book(s):

1. O. I. Elgerd, Electric Energy Systems Theory. New York, NY, USA: McGraw-Hill Education, 1995.
2. B. M. Weedy, B. J. Cory, N. Jenkins, J. Ekanayake, and G. Strbac, Electric Power Systems, 5th ed. Hoboken, NJ, USA: Wiley, 2012.
3. R. D. Bergen and V. Vittal, Power System Analysis. Upper Saddle River, NJ, USA: Pearson Education Inc., 1999.
4. M. L. Soni, P. V. Gupta, U. S. Bhatnagar, & A. Chakrabarti, "A Textbook on Power System Engineering", Dhanpat Rai & Co., 1998.
5. VIRTUAL LAB: <https://vp-dei.vlabs.ac.in/Dreamweaver/>

## EEPE3006 POWER STATION ENGINEERING (3-0-0)

**Course Learning Objectives:** This course will enable students to:

- To understand the student present day energy demand.
- To make the student to aware of components of power plants that run using conventional and non- conventional methods, factors affecting the site selection for a power plant and concept of base load plant and peak load plant.
- To expose the students to the principles of safety and environmental issues.
- To enable the student to recognize the importance of secondary energy source.

### Module I (05 Hours)

Introduction to different sources of energy and general discussion on their application to generation, Indian Energy Scenario.

Prediction of Load: Connected Load, Maximum Load, Demand Factor, Average load, Load Factor, Load duration curves, Diversity Factor, Choice of Type of Generation, Capacity Factor, Reserve Factor, Plant Use Factor, Base Load, Intermediate Load and Peak Load Plants.

Economics of power generation: Cost of Electrical Energy, Construction costs, Fixed cost, Costs for Energy, Depreciation of Plant, Fuel cost, Economic scheduling principle, Annual Operating Costs, Effect of Load Factor on cost per kWh, Tariff or Charge to Consumer.

### Module II (08 Hours)

Hydro electric power station: Selection of site for hydro-electric power plant. Hydrological cycle, precipitation, run-off and its measurement, hydrograph, flow duration and mass curves, Estimation of amount stored by a dam across the river, Storage and Pondage, Elementary idea about Earthen and Concrete Dam. Operational principle of Kaplan and Francis Turbine and Pelton wheel, Speed and Pressure Regulation, Work done and Arrangement and location of Hydroelectric Station: Catchment area, Reservoir, Dam, Head Gate, Spillways, Pen stock, Surge Tanks, Scroll case, Draft tubes and Tail Race, Power House, Classification of Hydroelectric Power Plants. Governors, Plant auxiliaries.

### Module III (08 Hours)

Thermal power station: Selection of site for thermal power plant Main Parts and Working of a Steam Station: Overall Block Diagram indicating the air circuit, coal and ash circuit, water and steam circuit, various types of steam turbines, ash and coal handling system, High Pressure and High-capacity water tube boilers, Economizer, Superheaters, De-Superheater, Re-heater, Air Pre-heater. Natural, Induced Forced and Balance Draft, PA fan, FD fan, ID fan, Chimney. Condensers, Feed water heaters, Evaporators, Make-up water, bleeding of steam, cooling water system. Electrostatic Precipitator: Basic working Principle and constructional details Governors, Plant auxiliaries.

### Module IV (06 Hours)

Nuclear power station: Introduction to fission & fusion, Principle of Nuclear Energy, Reactor Construction, Controlled Chain Reaction, Brief study of various Types of Power Reactor, Operational Control of Reactors, Location and layout of nuclear power plant, Economics of Nuclear Power Station.

### Module V (03 Hours)

Non Conventional Power Plants: Introduction to non conventional power plants (Solar, wind, geothermal, tidal, biogas) etc.

**Course Outcomes:** On completion of this course, students are able to:

- CO1: Present a sound knowledge about performance parameters in power plant economics and able to carry out tariff calculations.
- CO2: Summarize the general layout, components and operation of different conventional and non-conventional power plant.
- CO3: Describe various control systems of different power plants.

CO4: Describe energy conversion in power plants.

CO5: Realise the detrimental effects of power plants, causes of pollution and know about the preventive measures to be carried out to have a sustainable environment.

**Text Book(s):**

1. Power plant Engineering, P.K. Nag, TMH, 3rd Edition, 2013.
2. A Text Book of Power Plant Engineering, Rajput, Laxmi Publications, 4th edition, 2012

**Reference Book(s):**

1. Power plant engineering , P.C. Sharma, S.K., Kataria Publications, 2012.
2. M. V. Deshpande, "Elements of Electrical Power Station Design", PHI
3. Bernhardt G. A. Skrotzki, William A. Vopat, :Power Station Engineering and Economy", 2nd Edition, Tata McGraw Hill Publication.
4. Arora & Domkundwar , "A Course in Power Plant Engineering", Dhanpat Rai and sons.

## EEPE3009 POWER SYSTEM PROTECTION (3-0-0)

**Course Learning Objectives:** This course will enable students to:

1. To understand the significance of protection, protection schemes and role of earthing.
2. To study the characteristics, functions and application areas of various relays.
3. To acquire practical knowledge about common faults in power system apparatus and applying suitable protective schemes.
4. To understand the functioning of static relays and Numerical protection concepts.
5. To understand the problems associated with circuit breaking and to discuss about various circuit breakers.

### Module I (5 Hours)

Protection Schemes: Significance and need for protective schemes – nature and causes of faults – types of faults  
Effects of faults - Zones of protection and essential qualities of protection – Types of Protection schemes - Power system Grounding and Methods of Grounding.

### Module II (8 Hours)

Basics of Relays: Operating Principles and Relay Construction: Relay design and construction, Relay classification, Types of Electromagnetic relays, Theory of Induction relay torque, General Equations of Comparators and Electromagnetic Relays, Over Current relays, Directional relays, Distance relays, Differential relays. Feeder Protection: Over current, Distance and Pilot Protection. Static Relays: (Comparators and different relays) Amplitude comparator, Phase Comparator, Coincidence type phase comparator, Basic elements of a static relay, Over Current Relays, Differential Protection, Static distance Protection.

### Module III (5 Hours)

Apparatus Protection: Transformer Protection, Generator Protection, Motor Protection, Bus bar protection schemes.

### Module IV (6 Hours)

Numerical relays: Block Diagram of Numerical Relay, Signal Sampling & Processing, Numerical Over-current protection, Numerical Transformer differential Protection, Numerical distance Protection of Transmission Line.

### Module V (6 Hours)

Circuit Breaker: Physics of arcing phenomenon and arc interruption – DC and AC circuit breaking – re-striking voltage and recovery voltage - rate of rise of recovery voltage - current chopping - interruption of capacitive current - resistance switching - Types of circuit breakers – air blast, oil, SF<sub>6</sub> and vacuum circuit breakers – comparison of different circuit breakers – HVDC Breaker.

**Course Outcomes (CO):** On completion of this course, students are able to:

- CO1 Understand and select proper protective scheme and type of earthing.
- CO2 Explain the operating principles of various relays.
- CO3 Suggest suitable protective scheme for the protection of various power system apparatus.
- CO4 Analyze the importance of static relays and numerical relays in power system protection.
- CO5 Summarize the merits and demerits and application areas of various circuit breakers.

### Text Book(s):

1. Power System Protection and Switchgear – B.Ravindranath & M.Chander–New Age International Publishers (Second Edition)
2. Fundamentals of Power System Protection – Y.G.Paithankar and S.R.Bhide, PHI Publication.(Second Edition)
3. Electrical Power System - C.L.Wadhwa New Age International Publishers. (Sixth Edition).
4. Power System Engineering - M.L.Soni, P.V.Gupta, U.S.Bhatnagar, A.Chakrabarti, Dhanpat Rai & Co. (P) Ltd.
5. Protection and Switchgear - B.Bhalja, R.P.Maheshwari, N.G. Chothani, OXFORD University Press.
6. Power System Protection and Switchgear - Badri Ram, Vishwakarma, Tata McGraw hill.

**Reference Book(s):**

1. Power System relaying by Horwitz, Phadke, Research Press.
2. Arun Ingole, 'Switch Gear and Protection' Pearson Education, 2018.
3. Sunil S.Rao, 'Switchgear and Protection', Khanna Publishers, New Delhi, Four Edition, 2010.
4. NPTEL - <https://nptel.ac.in/courses/108/105/108105167/>



## HSHS3003 PROFESSIONAL ETHICS (3-0-0)

### Course Objectives :

This course aims to develop students' understanding of ethical principles, moral reasoning, and professional responsibilities. It introduces ethical theories, value systems, and the role of ethics in engineering and global contexts. The course prepares students to identify, analyze, and resolve ethical dilemmas in professional and societal scenarios with integrity.

### MODULE-I (6 Hours)

Introduction to Ethics:

Basic terms- Moral, Ethics, Ethical dilemma, Emotional intelligence

Moral development theories of Kohlberg and Piaget

View on ethics by Aristotle

Governing factors of an individual's value system

Personal and professional ethics

### MODULE-II (6 Hours)

Profession and Professionalism:

Clarification of the concepts: Profession, Professional, Professionalism, Professional accountability, Professional risks, Profession and Craftsmanship, Conflict of interest, Distinguishing features of a professional, Role and responsibilities of professionals, Professionals' duties towards the organization and vice-a-versa, Ethical Theories: Various ethical theories and their application- Consequentialism, Deontology, Virtue theory, Rights Theory, Casuist theory

Ethical terms: Moral absolutism, Moral Relativism, Moral Pluralism etc.; Resolving Ethical Dilemma

### MODULE-III (6 Hours)

Ethics in Engineering:

Purpose and concept of Engineering Ethics

Engineering as social experimentation

Types of inquiry

Issues in engineering ethics

Engineers' Responsibility and Safety:

Safety, Risk, Understanding the risk, Over estimating the risk, Risk-benefit analysis

Causes of an accident and identification of the preventive measures to be taken

Case Studies

### MODULE-IV (6 Hours)

Global Ethical Issues:

Different ethical issues in business, environment, IT, Bioethics, Intellectual Property Rights (IPR), Research, Media, CSR etc.

Ethical Codes:

Meaning and the significance of ethical codes

The limitations of ethical codes.

### Course Outcome

CO1: Define key ethical terms

CO2: Identify factors influencing value systems

CO3: Describe professional concepts

CO4: Explain the purpose of engineering ethics

### BOOKS FOR REFERENCE:

1. R. Subramanian, "Professional Ethics", Oxford University Press
2. Mike W. Martin and Roland Schinzinger, "Ethics in Engineering", Tata McGraw Hill
3. Charles E Harris, Michael S Pritchard and Michael J Rabins, "Engineering Ethics - Concepts and Cases", Thompson Learning
4. Daniel Albuquerque, "Business Ethics", Oxford University Press
5. Edmund G. Seebauer and Robert L. Barry, "Fundamentals of Ethics", Oxford University Press

## **MCMC3001 ENVIRONMENTAL ENGINEERING (3-0-0)**

### **Course Objectives:**

- To acquire basic knowledge of source of water and various treatment processes
- To determine the sewage quantity, and understand its treatment and disposal
- To Identify and value the effect of the pollutants in atmosphere
- To formulate strategies to solid waste management

### **Module-I: (08 Hrs)**

Water: Sources of Water and quality issues, water quality requirement for different beneficial uses, Water quality standards, water quality indices, water safety plans, Water Supply systems, Need for planned water supply schemes, Water demand industrial and agricultural water requirements, Components of water supply system; Transmission of water, Distribution system, Various valves used in W/S systems, service reservoirs and design. Water Treatment: aeration, sedimentation, coagulation flocculation, filtration, disinfection, advanced treatments like adsorption, ion exchange, membrane processes.

### **Module-II: (08 Hrs)**

Sewage- Domestic and Storm water, Quantity of Sewage, Sewage flow variations. Conveyance of sewage- Sewers, shapes design parameters, operation and maintenance of sewers, Sewage pumping; Sewerage, Sewer appurtenances, Design of sewerage systems. Small bore systems, Storm Water- Quantification and design of Storm water; Sewage and Sullage, Pollution due to improper disposal of sewage, National River cleaning plans, Wastewater treatment, aerobic and anaerobic treatment systems, suspended and attached growth systems, recycling of sewage – quality requirements for various purposes.

### **Module-III: (08 Hrs)**

Air - Composition and properties of air, Quantification of air pollutants, monitoring of air pollutants, Air pollution- Occupational hazards, Urban air pollution automobile pollution, Chemistry of combustion, Automobile engines, quality of fuel, operating conditions and interrelationship. Air quality standards, Control measures for Air pollution, construction and limitations

### **Module-IV: (08 Hrs)**

Noise-Basic concept, measurement and various control methods. Solid waste Management-Municipal solid waste, Composition and various chemical and physical parameters of MSW, MSW management: Collection, transport, treatment and disposal of MSW. Special MSW: waste from commercial establishments and other urban areas, solid waste from construction activities, biomedical wastes, Effects of solid waste on environment: effects on air, soil, water surface and ground health hazards. Disposal of solid waste-segregation, reduction at source, recovery and recycle. Disposal methods- Integrated solid waste management. Hazardous waste: Types and nature of hazardous waste as per the HW Schedules of regulating authorities.

### **Course Outcomes:**

After successfully studying this course, students will able to:

- Understand the impact of humans on environment and environment on humans
- Identify and value the effect of the pollutants on the environment: atmosphere, water and soil
- Formulate strategies to control, reduce and monitor pollution
- Determine the most appropriate technique for the treatment of water, wastewater solid waste and contaminated air

### **Books**

- Introduction to Environmental Engineering and Science by Gilbert Masters, Prentice Hall, New Jersey.
- Introduction to Environmental Engineering by P. Aarne Vesilind, Susan M. Morgan, Thompson /Brooks/Cole; Second Edition 2008.
- Peavy, H.s, Rowe, D.R, Tchobanoglous, G. Environmental Engineering, Mc-Graw -Hill International Editions, New York 1985.
- MetCalf and Eddy. Wastewater Engineering, Treatment, Disposal and Reuse, Tata McGraw-Hill, New Delhi

## **CSPS3202 PROJECT FOR PRODUCT DEVELOPMENT - I (0-0-6)**

### **Course objectives:**

Learning outcomes Upon completing the course, students are expected to be able to:

- Plan and execute independently projects aimed at collecting, systematize and analyze information about markets and customer contexts as fundamental supporting elements for product development in specific sectors and industrial fields,
- Apply the most important models for organizing and managing product development and its implementation in concrete commercial setting,
- Analyze complex product development situations and, based on such analysis, suggest relevant strategies, plans and action programs for various types of companies and organizations,
- Identify the need of further knowledge and tools (both analytics and computer-based) to conduct product development tasks,
- Evaluate critically the result of a product development project and, based on such an evaluation, reflect on uncertainty and risks in execution, so to be able to suggest alternative conclusions.

### **Instruction**

The course is organized as an independent project conducted in teams of 4-5 students with the aim of developing an idea all the way to a product ready for launch for a specific firm. This work is supported by methodological lectures. The project work is presented during a series of seminars with oppositions acting as "control gates" and leading to a final seminar where a decision on launching or not the product is taken by an opponent group.

## EOPC3002 DIGITAL SIGNAL PROCESSING (3-0-0)

### Course Objectives:

This course introduces fundamental concepts of digital signal processing, covering signal representation, z-transforms, and discrete-time systems. Students will analyze LTI systems using DFT/FFT, design FIR/IIR filters (window method, bilinear transformation), and implement adaptive filters. Emphasis is placed on practical applications, including spectral estimation and digital filter realization.

### Module-I: (06 hours)

**Signals:** Representation of signals on orthogonal basis, sampling and reconstruction of signals, Discrete time signals/sequences, Discrete time systems. Analysis and response (convolution sum) of discrete - time linear LTI system, Correlation of discrete time Signal.

The z-transform, Analysis of LTI systems using z-transform, Properties of z-transform.

### Module-II: (06 hours)

Inverse Z-Transform, Inversion Z-Transform by Power Series Expansion, Inversion of the Z-Transform by Partial-Fraction Expansion, Analysis of Linear Time-Invariant Systems in the z-Domain.

Frequency analysis of LTI systems: Discrete Fourier transform (DFT), frequency domain sampling, Properties of DFT, Frequency

### Module-III: (06 hours)

Efficient computation of DFT: circular convolution, circular correlation, linear filtering methods based on DFT.

Fast Fourier transform (FFT): Decimation in time (DIT) algorithm, Decimation in frequency (DIF) algorithm, Application of FFT.

### Module-IV: (08 hours)

Realization of FIR and IIR systems using direct forms and cascaded forms. Design of Digital filters: General considerations. Design of FIR filters: window method.

Design of IIR filters: Impulse invariance method, bilinear transformation method for analog filters.

### Module-V: (04 hours)

Basic adaptive filter: Structure of Adaptive FIR filter, System Modeling and Inverse Modelling. Application of DSP.

### Course Outcome:

Upon completion of the course, the students will be able to:

CO1: Analyze and characterize signals and systems.

CO2: Analyze digital systems in time and frequency domain.

CO3: Demonstrate digital system characterization through DFT and FFT.

CO4: Implement digital filters and systems.

CO5: Demonstrate adaptive signal spectral estimation methods

### Text Books:

1. Digital Signal Processing – Principles, Algorithms and Applications by J. G. Proakis and D. G. Manolakis, Pearson Education, Samuel Y. Liao, Pearson Education
2. Digital Signal Processing - Dr. Shalia D. Apte, Willey Publication
3. Digital Signal Processing: Tarun Kumar Rawat, Oxford University Press.
4. Digital Signal Processing – S. Salivahan, A. Vallavraj and C. Gnanapriya, Tata McGrawHill.
5. Digital Signal Processing – Manson H. Hayes (Schaum's Outlines) Adapted by Subrata Bhattacharya, Tata McGraw Hill.

## **EOPC3202 DIGITAL SIGNAL PROCESSING LABORATORY (0-0-3)**

### **Course Objectives:**

This laboratory course provides hands-on experience in digital signal processing using MATLAB and DSP kits. Students will generate waveforms, implement convolution (linear/circular), compute FFT, and design FIR/IIR filters. Emphasis is placed on spectral analysis, noise cancellation, and practical implementation of DSP algorithms to analyze and process real-world signals.

### **(Any Ten of the following experiments are to be performed)**

1. Familiarization with the architecture of a standard DSP kit.
2. Generation of various types of waveforms (sine, cosine, square, triangular etc.) using MATLAB and DSP kit.
3. Linear convolution of sequences (without using the inbuilt conv. function in MATLAB) and verification of linear convolution using DSP kit.
4. Circular convolution of two sequences and comparison of the result with the result obtained from linear convolution using MATLAB and DSP kit.
5. (i) Computation of autocorrelation of a sequence, cross correlation of two sequences using MATLAB.  
(ii) Computation of the power spectral density of a sequence using MATLAB, also implementing the same in a DSP kit.
6. Finding the convolution of a periodic sequence using DFT and IDFT in MATLAB.
7. (i) Implementation of FFT algorithm by decimation in time and decimation in frequency using MATLAB.  
(ii) Determine and plot the FFT of a given 1-D signal using DSP kit. Scattering parameters of Circulator /Isolators.
8. Design and implementation of FIR (lowpass and highpass) Filters using windowing techniques (rectangular window, triangular window and Kaiser window) in MATLAB and DSP kit.
9. Design and implementation of IIR (lowpass and highpass) Filters (Butterworth and Chebyshev) in MATLAB and DSP kit.
10. Convolution of long duration sequences using overlap add, overlap save using MATLAB.
11. Implementation of noise cancellation using adaptive filters on a DSP kit.

### **Course Outcomes:**

Upon completion of the subject, the students will demonstrate the ability to:

- CO1: Analyze different types of transformations.
- CO2: Implement and modify of LTI systems.
- CO3: Analyze the performance of DIT & DIF algorithms.
- CO4: Implement IIR & FIR systems.
- CO5: Demonstrate conclusions after computing PSD.